

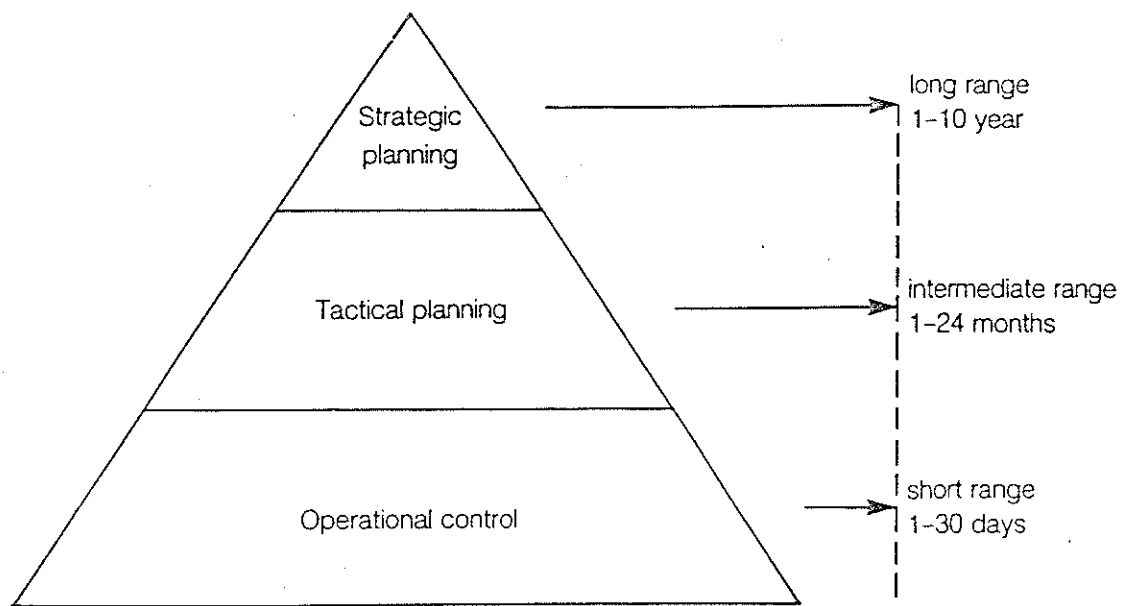


Figure 3.14 The hierarchies of planning.

plans for each functional area over a period of several months. The tactical production plan would establish a master schedule for production of individual end items, establish resulting shop order release dates, and set levels of machine capacity, tooling, labour and maintainance. The operational control level would be concerned with the detailed execution of tactical plans on a weekly or daily basis, for each work centre or cell. Individual functions by level are shown in more detail in Figure 3.15.

Interactions between plans at a given level are weak, and take place through the constraints imposed by the plan at the next higher level. Planners at each level have autonomy within these constraints to accommodate local contingencies as they see fit. As shown in Figure 3.14, the main feedback loops in this process are from level to level. Progress with implementing a plan at one level is reported to the level above, which is responsible for modifying its own plan, if necessary, on the basis of the information received. It is here that most of the practical problems arise. In a functionally divided organisation, modification of a higher level plan due to a contingency within a particular lower-level subsystem can have implications for all other subsystems. Frequently occurring contingencies can therefore result in all subsystems having continuously to modify their own plans because of altered constraints propagated from above. This can put such a severe information processing load on individual subsystems and inter-level communication channels, that existing vertical information systems break down, as so often appears to happen in the case of MRP. The solution generally employed is to build additional slack into plans so that more contingencies can be accommodated at a given level before necessitating replanning at the higher level.

<i>Planning level</i>	<i>Functions</i>
Strategic planning	Product design
	Aggregate resource management
	Aggregate forecasting
	New product planning
	Long lead time ordering
	Manufacturing engineering
	Systems design
Tactical planning	Master production scheduling
	Due-date setting
	Requirements planning
	Capacity planning
	Purchasing
	Aggregate inventory management
Operational control	Preventative maintenance planning
	Shop order release
	Shop floor scheduling
	Dispatching
	Receiving
	Capacity control
	Tool control
	Inventory control
	Diagnostic maintenance
	Quality control
	Cost control

Figure 3.15 Planning and control functions by level.

3.3.4 Investment in lateral communication systems

Lateral communication systems can be used as an alternative method to resolve the effects of such contingencies. These often take the form of informal relationships between subsystems (for example, the insidious informal system that often runs in parallel with formal MRP systems). These can cause difficulties when used in conjunction with formal hierarchical planning based on a vertical information system, if they result in actions of which the formal system is not aware. Provision for formalised lateral communication is not generally provided as part of a